

SIXTH SEMESTER U.G. DEGREE EXAMINATION, MARCH 2022**(CBCSS-UG)****Mathematics****MTS 6B 13—DIFFERENTIAL EQUATIONS****(2019 Admissions)****(Multiple Choice Questions for SDE Candidates)****Time : 15 Minutes****Total No. of Questions : 20****Maximum : 20 Marks****INSTRUCTIONS TO THE CANDIDATE**

1. This Question Paper carries Multiple Choice Questions from 1 to 20.
2. The candidate should check that the question paper supplied to him/her contains all the 20 questions in serial order.
3. Each question is provided with choices (A), (B), (C) and (D) having one correct answer. Choose the correct answer and enter it in the main answer-book.
4. The MCQ question paper will be supplied after the completion of the descriptive examination.

11. $L\left[t^{-1/2}\right] = \underline{\hspace{2cm}}.$

(A) $-\sqrt{\frac{\pi}{2s}}.$

(B) $\sqrt{\frac{\pi}{2s}}.$

(C) $-\sqrt{\frac{\pi}{s}}.$

(D) $\sqrt{\frac{\pi}{s}}.$

12. The inverse Laplace transform of $\frac{2\pi}{s+\pi}$ is $\underline{\hspace{2cm}}.$

(A) $-4\pi e^{-\pi t}.$

(B) $4\pi e^{-\pi t}.$

(C) $-2\pi e^{-\pi t}.$

(D) $2\pi e^{-\pi t}.$

13. $L^{-1}\left[\frac{1}{s^4}\right] = \underline{\hspace{2cm}}.$

(A) $\frac{t^3}{6}.$

(B) $\frac{t^3}{4}.$

(C) $-\frac{t^3}{6}.$

(D) $\frac{t^3}{10}.$

14. $L^{-1}\left[\frac{s}{s^2+36}\right] = \underline{\hspace{2cm}}.$

(A) $-\cos 6t.$

(B) $\cos 6t.$

(C) $\cos 3t.$

(D) $-\cos 3t.$

15. $L^{-1}\left[\frac{5s}{s^2+36}\right] = \underline{\hspace{2cm}}.$

(A) $-5 \cos 6t.$

(B) $5 \cos 6t.$

(C) $-5 \sin 6t.$

(D) $5 \sin 6t.$

16. The inverse Laplace transform of $\frac{2}{s^2+16}$ is $\underline{\hspace{2cm}}.$

(A) $\frac{\sin 5t}{4}.$

(B) $\frac{\sin 4t}{5}.$

(C) $-\frac{\sin 4t}{4}.$

(D) $\frac{\sin 4t}{4}.$

17. The inverse Laplace transform of $\frac{1}{s(s+1)(s+2)}$ is _____.

(A) $-\frac{1}{2}e^{-t} + \frac{1}{2}e^{-2t}$.

(B) $\frac{1}{2}e^{-t} + \frac{1}{2}e^{-2t}$.

(C) $\frac{1}{2}e^{-t} - \frac{1}{2}e^{-2t}$.

(D) None of the above options.

18. $L^{-1}\left[\frac{3s+1}{(s-1)(s^2+1)}\right] = \text{_____}.$

(A) $2e^t - 2\cos t + \sin t$.

(B) $-2e^t - 2\cos t + \sin t$.

(C) $2e^t - 2\cos t - \sin t$.

(D) $2e^t + 2\cos t + \sin t$.

19. The inverse transform of $\frac{s+2}{(s+2)^2+1}$ is _____.

(A) $e^{-2t} \cos t$.

(B) $e^{2t} \cos t$.

(C) $e^{-2t} \sin t$.

(D) $e^{2t} \sin t$.

20. $L[t \sin t] = \text{_____}.$

(A) $\frac{2s}{(s^2+1)^2}$.

(B) $\frac{2s}{(s^2-1)^2}$.

(C) $-\frac{2s}{(s^2+1)^2}$.

(D) None of the above options.

SIXTH SEMESTER U.G. DEGREE EXAMINATION, MARCH 2022

(CBCSS-UG)

Mathematics

MTS 6B 13—DIFFERENTIAL EQUATIONS

(2019 Admissions)

Time : Two Hours and a Half

Maximum : 80 Marks

Section A

*Answer at least ten questions.**Each question carries 3 marks.**All questions can be attended.**Overall Ceiling 30.*

1. Find the general solution of the differential equation $\frac{dy}{dt} = -ay + b$ where a,b are positive real numbers.
2. Determine the values of r for which e^{rt} is a solution of the differential equation $y''' - 3y'' + 2y' = 0$.
3. Using method of integrating factors solve the differential equation $\frac{dy}{dt} - 2y = 4 - t$.
4. Find the solution of the differential equation :

$$\frac{dy}{dx} = \frac{3x^2 + 4x + 2}{2(y-1)}, y(0) = -1.$$

5. Find the Wronskian of the functions $\cos^2 0, 1 + \cos(2\theta)$.
6. Find the general solution of the differential equation $y'' + 2y' + 2y = 0$.
7. Let $y = \phi(x)$ be a solution of the initial value problem :

$$(1+x^2)y'' + 2xy' + 4x^2y = 0, y(0) = 0, y'(0) = 1.$$

Determine $\phi'''(0)$.

8. Determine a lower bound for the radius of convergence of series solutions about each given point $x_0 = 4$ for the given differential equation $y'' + 4y' + 6xy = 0$.

Turn over

9. Find the Laplace transform of the function $\sin(at)$.
10. Find the inverse Laplace transform of $\frac{n!}{(s-a)^{n+1}}$ where $s > a$.
11. Let $u_c(t)$ be unit step function and $L(f(t)) = F(s)$. Show that :
- $$L(u_c(t)f(t-c)) = e^{-cs}F(s).$$
12. Find the inverse Laplace transform of the following function by using the convolution theorem
- $$\frac{1}{s^4(s^2+1)}.$$
13. Solve the boundary value problem :
- $$y'' + y = 0, y(0) = 0, y(\pi) = 0.$$
14. Define an even function and show that if $f(x)$ is an even function then :
- $$\int_{-L}^L f(x) dx = 2 \int_0^L f(x) dx.$$
15. Define the following partial differential equations :
- heat conduction equation.
 - one-dimensional wave equation.

(10 × 3 = 30 marks)

Section B*Answer at least five questions.**Each question carries 6 marks.**All questions can be attended.**Overall Ceiling 30.*

16. Let $y_1(t)$ be a solution of $y' + p(t)y = 0$ and let $y_2(t)$ be a solution of $y' + p(t)y = g(t)$.
Show that $y(t) = y_1(t) + y_2(t)$ is also a solution of equation $y' + p(t)y = g(t)$.
17. Find the value of b for which the following equation is exact, and then solve it using that value of b .
- $$(xy^2 + bx^2y) + (x+y)x^2y' = 0.$$
18. Solve the initial value problem
- $$y'' + 4y = t^2 + 3e^t, y(0) = 0, y'(0) = 2.$$

19. Use method of variation of parameters find the general solution of :

$$y'' + 4y = 8 \tan t, -\pi/2 < t < \pi/2.$$

20. Find the solution of the initial value problem :

$$2y'' + y' + 2y = \delta(t - 5), y(0) = 0, y'(0) = 0.$$

here $\delta(t)$ denote the unit impulse function.

21. Using Laplace transform solve the initial value problem :

$$y'' + 4y = 0, y(0) = 3, y'(0) = -1.$$

22. Find the co-efficients in the Fourier series for f :

$$f(x) = \begin{cases} 0, & -3 < x < -1 \\ 1, & -1 < x < 1 \\ 0, & 1 < x < 3 \end{cases}$$

Also suppose that $f(x + 6) = f(x)$.

23. Find the solution of the following heat conduction problem :

$$100u_{xx} = u_t, 0 < x < 1, t > 0$$

$$u(0, t) = 0, u(1, t) = 0, t > 0$$

$$u(x, 0) = \sin(2\pi x) - \sin(5\pi x), 0 \leq x \leq 1.$$

(5 × 6 = 30 marks)

Section C

*Answer any two questions.
Each question carries 10 marks.*

24. Find the general solution of the following differential equation using the method of integrating factors :

$$\frac{dy}{dt} + \frac{1}{2}y = \frac{1}{2}e^{t/3}.$$

Draw some representative integral curves of the differential equation and also find the particular solution whose graph contains the point (0,1).

25. Find a series solution of the differential equation :

$$y'' + y = 0, -\infty < x < \infty.$$

26. Find the Laplace transform of $\int_0^t \sin(t - \tau) \cos \tau \, d\tau$

27. Find the temperature $u(x, t)$ at any time in a metal rod 50 cm long, insulated on the sides, which initially has a uniform temperature of 20°C throughout and whose ends are maintained at 0°C for all $t > 0$.

(2 × 10 = 20 marks)